

Case 1: Dirichlet Boundary Condition Without Source

1. Problem Statement

The objective of this case study is to solve the one-dimensional transient diffusion equation using the Finite Volume Method (FVM) implemented in FiPy. The problem considers a one-dimensional domain subjected to Dirichlet boundary conditions at both ends and does not include any internal source term.

The computational domain extends from $x = 0$ to $x = 1$. Fixed values are prescribed at both boundaries, with $y = 0$ at the left boundary and $y = 1$ at the right boundary. The transient solution is evolved in time until steady state is reached.

The numerical solution obtained using FiPy is validated against the analytical steady-state solution. Additional studies including transient evolution, numerical–analytical comparison, mesh convergence, timestep convergence, and error analysis are performed to assess the accuracy and stability of the numerical implementation.

2. Governing Equation

The governing equation for one-dimensional transient diffusion without a source term is

$$\partial y / \partial t = k \partial^2 y / \partial x^2$$

where y is the dependent variable, k is the diffusion coefficient, x is the spatial coordinate, and t represents time.

For the present study:

- Diffusion coefficient, $k = 1$
- Domain length, $L = 1$

The boundary conditions are:

$$y(0,t) = 0$$

$$y(1,t) = 1$$

At steady state:

$$d^2y/dx^2 = 0$$

Integrating once:

$$dy/dx = C_1$$

Integrating again:

$$y = C_1x + C_2$$

Applying $y(0)=0$:

$$C_2 = 0$$

Applying $y(1)=1$:

$$C_1 = 1$$

Therefore,

$$y(x) = x$$

This analytical solution is used as the reference solution for validating the numerical results obtained using FiPy.

3. Numerical Implementation

The governing diffusion equation was solved numerically using the FiPy finite volume package in Python. A one-dimensional uniform mesh was generated over the domain $0 \leq x \leq 1$.

Simulation parameters:

Parameter	Value
Domain Length (L)	1.0
Number of Cells (nx)	50
Cell Size (Δx)	0.02
Diffusion Coefficient (k)	1.0
Time Step (Δt)	1×10^{-3}
Number of Time Steps	2000

The initial condition was

$$y(x,0) = \sin(\pi x)$$

Dirichlet boundary conditions were imposed at both ends:

$$y(0,t)=0$$

$$y(1,t)=1$$

The transient diffusion equation was solved iteratively until convergence to steady state was achieved.

4. Results and Discussion

4.1 Transient Evolution of the Solution

The transient diffusion equation was solved using FiPy starting from the initial condition $y(x,0)=\sin(\pi x)$. As the simulation progressed, the solution evolved under the influence of diffusion and gradually approached the steady-state profile imposed by the Dirichlet boundary conditions.

The transient evolution demonstrates the smoothing effect of diffusion, where the initial non-linear profile progressively transforms into the final linear steady-state solution.

The solution remains stable throughout the simulation and converges monotonically toward the steady-state configuration.

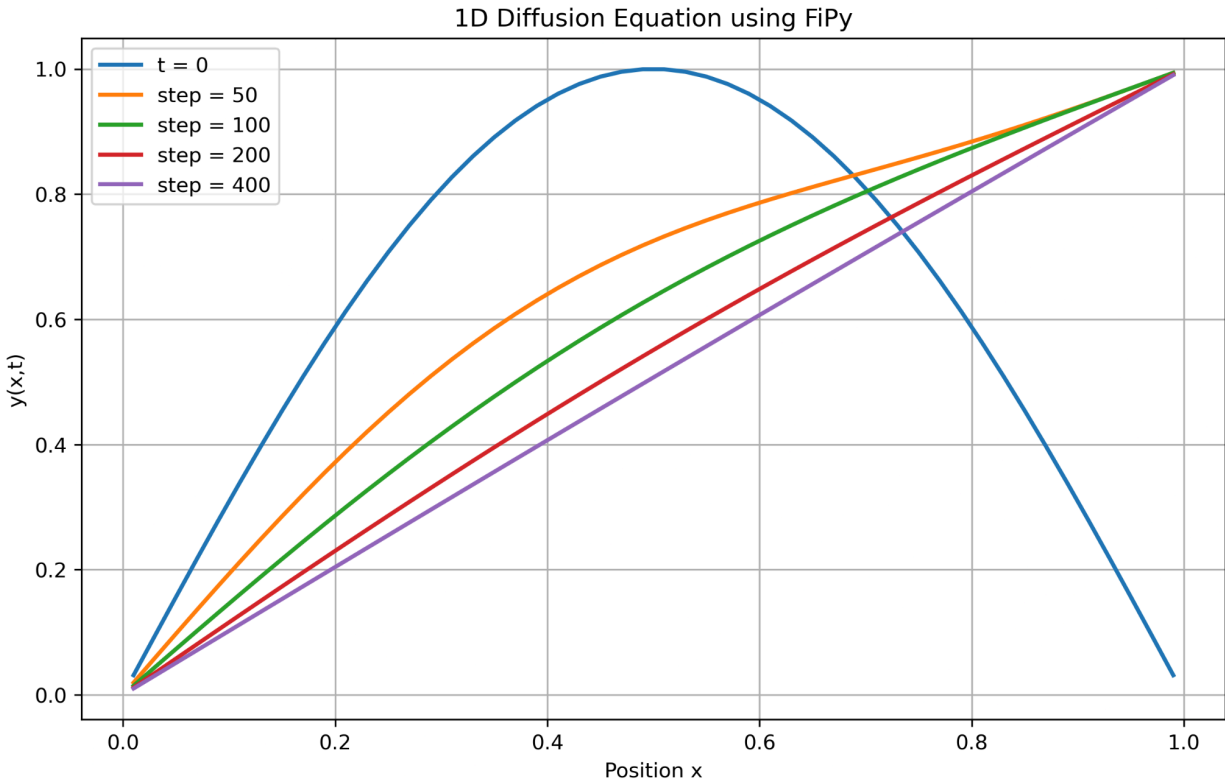


Figure 1. Transient evolution of the solution.

The solution gradually approaches the steady-state profile as time progresses.

4.2 Time-Step Convergence Study

A time-step convergence study was performed to evaluate the sensitivity of the transient solution to temporal discretization. Simulations were carried out using different time-step sizes while keeping all other numerical parameters unchanged.

The results indicate that as the time-step size decreases, the numerical solution converges toward a consistent solution profile. The differences between solutions obtained using smaller time steps become negligible, demonstrating temporal convergence of the numerical scheme.

The study confirms that the chosen time-step size provides a stable and accurate representation of the transient diffusion process while maintaining computational efficiency.

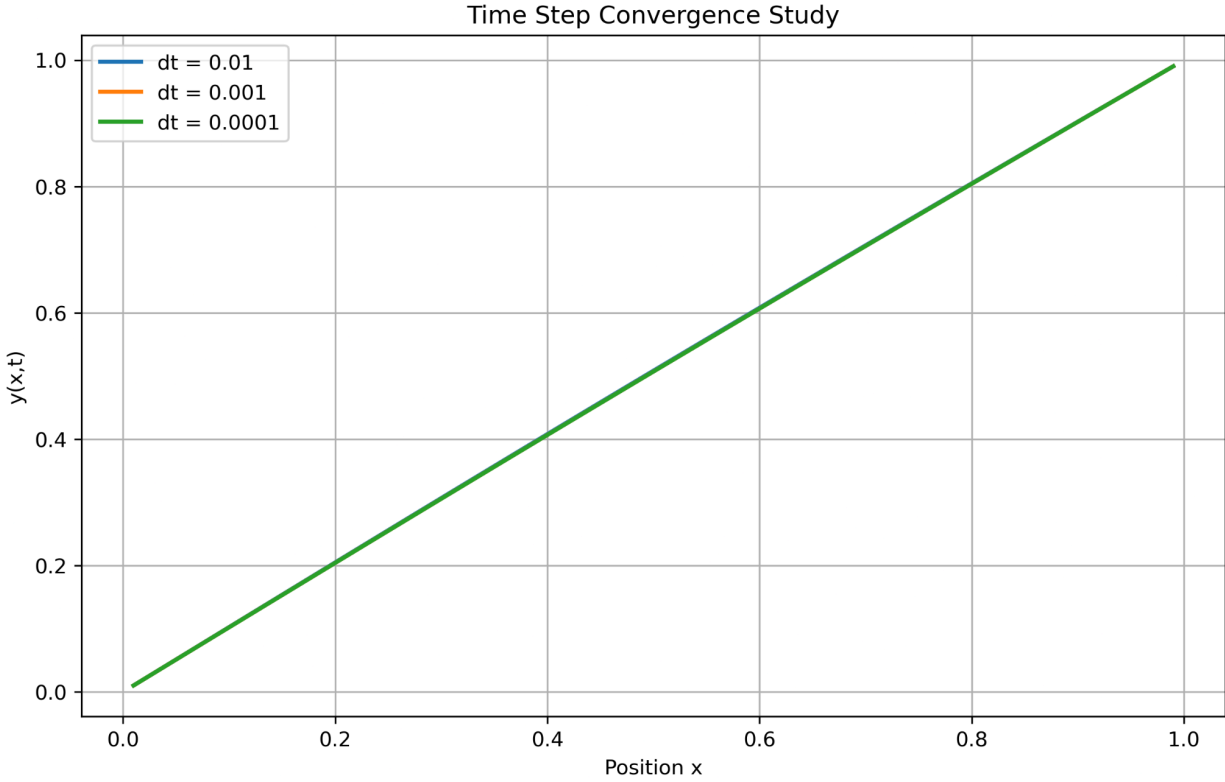


Figure 2. Time-step convergence study.

The solutions obtained using different time-step sizes show excellent agreement. This indicates that the numerical solution is largely independent of temporal discretization for sufficiently small time steps.

The overlap of the solutions obtained using $\Delta t = 10^{-2}$, 10^{-3} , and 10^{-4} indicates temporal independence of the numerical solution.

4.3 Mesh Convergence Study

A mesh convergence study was performed to investigate the influence of spatial discretization on the numerical solution. Simulations were conducted using progressively finer meshes while maintaining the same physical parameters and boundary conditions.

As the mesh density increased, the numerical solution approached a limiting profile corresponding to the analytical solution. The differences between successive mesh refinements decreased significantly, indicating convergence of the finite volume discretization.

The results demonstrate that the numerical solution becomes increasingly independent of the mesh size as the spatial resolution is refined.

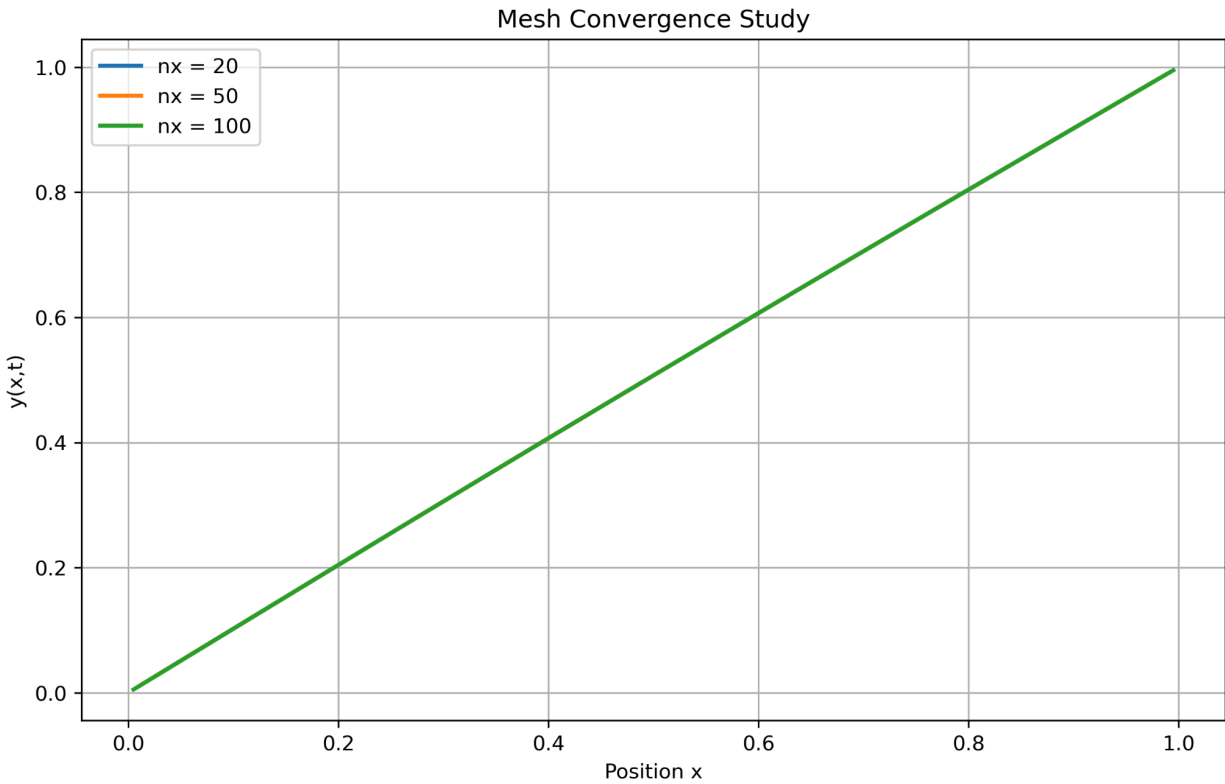


Figure 3. Mesh convergence study.

As the mesh is refined, the numerical solution converges toward a limiting profile. The decreasing difference between successive meshes confirms spatial convergence of the finite volume method.

The solutions obtained using $nx = 20$, 50 , and 100 cells are nearly indistinguishable, confirming spatial convergence.

4.4 Steady-State Solution

After a sufficient number of time steps, the transient solution converged to a steady-state profile. Under the prescribed Dirichlet boundary conditions, the analytical steady-state solution is

$$y(x)=x$$

The numerical solution obtained using FiPy successfully reproduced the expected linear behavior across the entire computational domain.

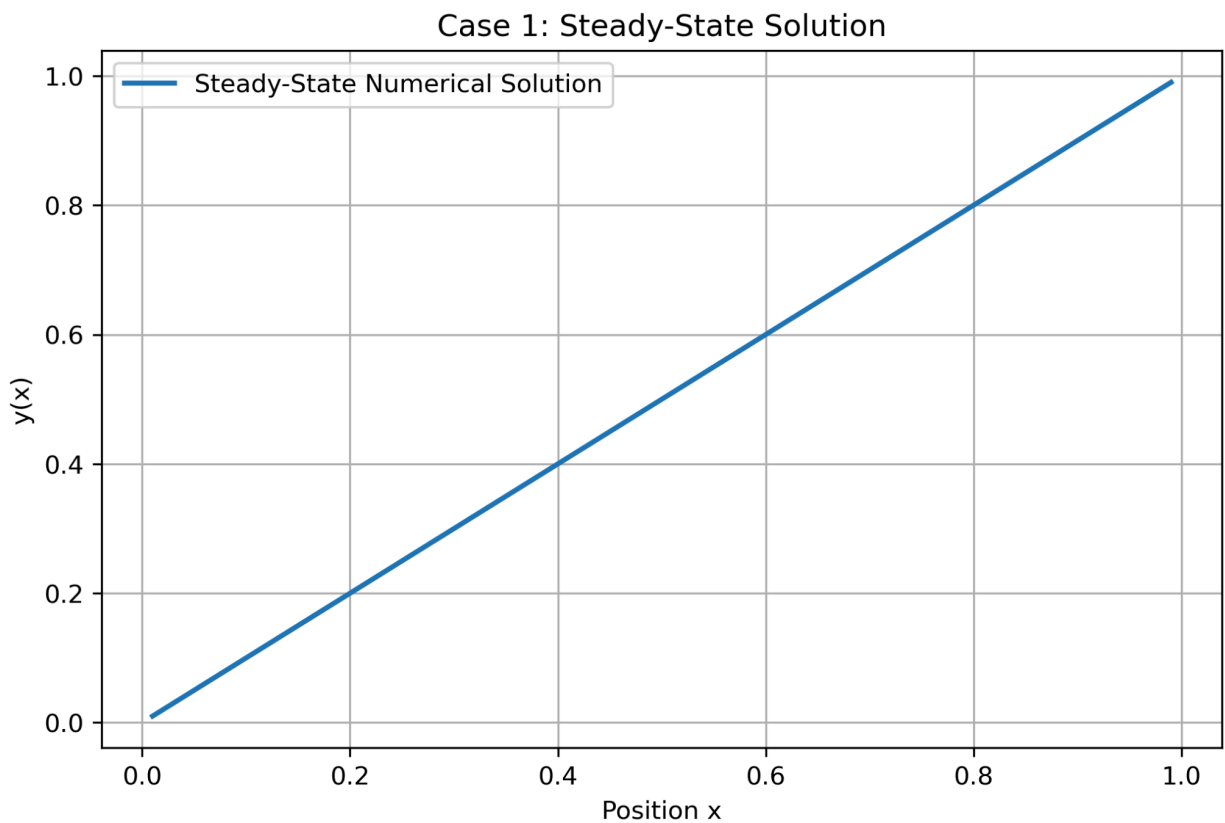


Figure 4. Steady-state solution obtained using FiPy.

The final numerical solution exhibits the expected linear profile across the domain. The solution satisfies the prescribed Dirichlet boundary conditions and represents the steady-state diffusion behavior.

4.5 Steady-State Verification

To verify that the solution had reached steady state, the maximum change in the solution between successive time steps was monitored throughout the simulation.

As the transient evolution progressed, the magnitude of these changes decreased steadily and eventually approached zero, confirming convergence to a stable equilibrium state.

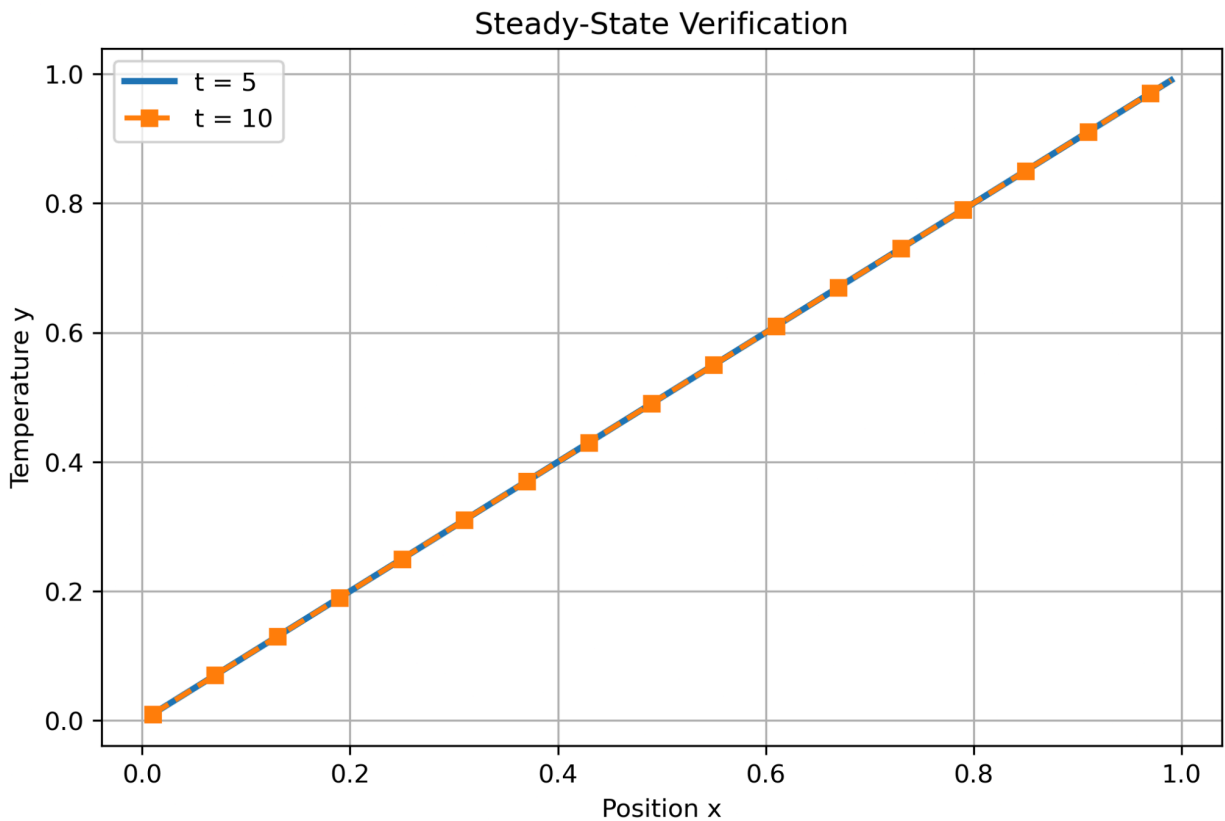


Figure 5. Verification of steady-state convergence.

The magnitude of changes between successive iterations approaches zero with time. This confirms that the numerical solution has reached a stable steady-state condition.

The profiles at $t = 5$ and $t = 10$ overlap almost completely, indicating that the solution has reached steady state.

4.6 Numerical vs Analytical Comparison

The steady-state numerical solution obtained using FiPy was compared directly with the analytical solution $y(x)=x$.

The comparison demonstrates excellent agreement between the numerical and analytical solutions throughout the computational domain. The numerical profile closely overlaps the analytical profile, confirming the correctness of the implementation.

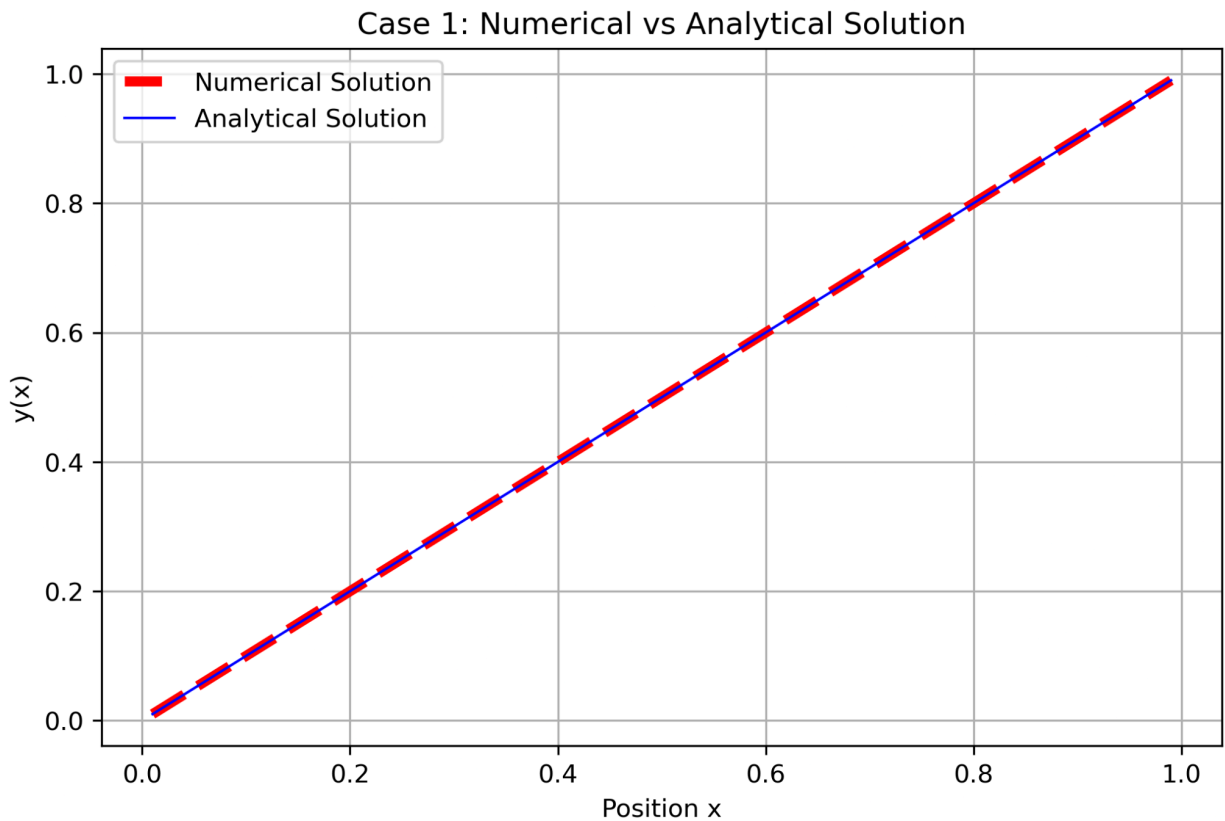


Figure 6. Comparison between numerical and analytical solutions.

The numerical and analytical solutions overlap almost perfectly throughout the computational domain. This demonstrates the accuracy of the finite volume implementation.

The analytical solution represents a continuous function over the domain, while the numerical solution is computed at discrete mesh locations. The close agreement between the two validates the finite volume implementation.

4.7 Error Analysis

To quantify the accuracy of the numerical solution, the difference between the numerical and analytical solutions was evaluated across the computational domain.

The maximum error obtained between the numerical and analytical solutions was

$$\text{Maximum Error} = 4.53 \times 10^{-14}$$

while the L2 error norm was

$$\text{L2 Error} = 3.25 \times 10^{-14}.$$

These error values are extremely small compared to the magnitude of the solution, indicating excellent agreement between the numerical and analytical results. The observed discrepancy arises primarily from numerical discretization and floating-point precision effects.

The maximum error is of the order of machine precision, demonstrating near-perfect agreement between the numerical and analytical solutions.

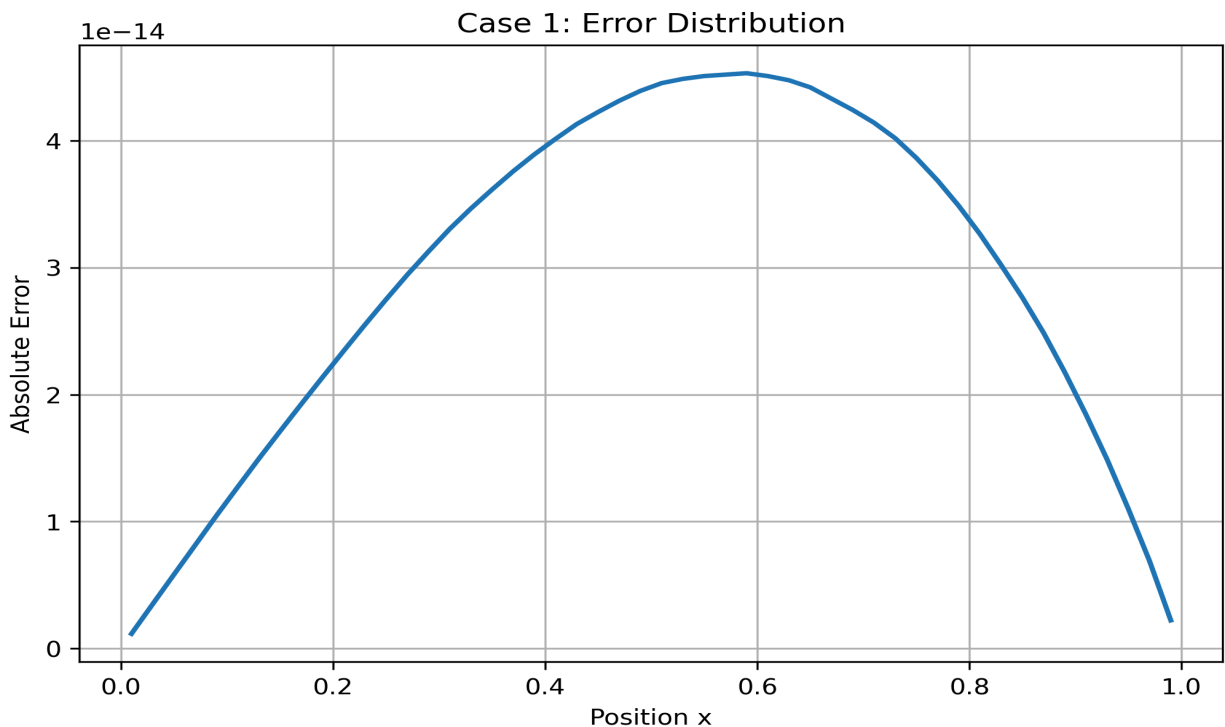


Figure 7. Error distribution between numerical and analytical solutions.

The error remains extremely small across the domain, indicating excellent agreement between the numerical and analytical solutions. The observed deviations are primarily due to discretization and numerical precision effects.

The maximum error is of the order of 10^{-14} , indicating near machine-precision agreement between the numerical and analytical solutions.

5. Conclusion

In this case study, the one-dimensional transient diffusion equation without a source term was successfully solved using the Finite Volume Method implemented in FiPy.

The numerical results exhibited excellent agreement with the analytical steady-state solution. Time-step convergence and mesh convergence studies confirmed the numerical stability and consistency of the solution, while error analysis demonstrated negligible deviation from the analytical result.

Overall, the study verifies the accuracy of the finite volume approach for solving one-dimensional diffusion problems and establishes a foundation for the subsequent investigation of more complex boundary conditions and source terms.